

Home Lab — Day 1: On-Prem Active Directory

Self-directed home lab to practise Level 1 service desk / IT support identity tasks. Not paid or production work — built for learning and interview preparation.

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Environment

Host	Windows host, 16 GB RAM, Oracle VirtualBox
Server VM	DC01 — Windows Server 2022 Standard (Evaluation), Desktop Experience. 4 GB RAM, 2 vCPU, 60 GB dynamic disk
Network	VirtualBox NAT Network. Static IP 10.0.2.15 / 255.255.255.0, gateway 10.0.2.2, DNS pointed to self (127.0.0.1)
Domain	New forest corp.lab.local (NetBIOS: CORP)

Tasks completed (core L1 identity work)

Task	What I did	Why it matters
Build domain controller	Installed Server 2022 from scratch, set static IP, renamed to DC01, installed AD DS and promoted to a Domain Controller in a new forest. DNS installed automatically.	A DC needs a fixed IP and self-DNS before promotion; renaming after is messy.
Organise the directory	Built an OU tree: CORP > Users, Groups, Computers, Departments > IT, Sales. Kept accidental-deletion protection on.	Real environments use OUs to organise objects and target Group Policy, not the default Users folder.
Create a user	Created alex.taylor in the IT OU using firstname.lastname, with 'must change password at next logon'.	Mirrors the standard 'new starter account' ticket; the user sets their own secret on first login.
Password reset	Reset the user's password with a temporary password + force-change + unlock option.	The single most common L1 ticket.
Lockout policy	Set account lockout threshold to 3 attempts via Default Domain Policy.	Makes lockouts behave realistically; basis for handling 'I'm locked out' tickets.
Group-based access	Created Global Security group IT-Staff and added the user as a member.	Access is granted via group membership, not per-user — standard practice.

Skills demonstrated

Active Directory Domain Services • DNS fundamentals • Windows Server 2022 setup • static IP configuration • OU design • user lifecycle (create / reset / unlock) • Group Policy (account lockout) • security groups • VirtualBox virtualisation.

Next session

Day 1b — build a Windows 11 Pro client VM, join it to corp.lab.local, and log in as the test user to make account lockouts and Group Policy observable end-to-end. Then Day 2: Microsoft 365 Developer tenant + Entra ID, repeating the same identity tasks in the cloud.

Evidence: screenshots captured at each milestone (ipconfig, static IP, promotion config, CORP\ login, OU tree, new user, reset dialog, lockout policy, group membership). Intended for a self-directed home-lab GitHub repository.